

REPORT_20

This report will talk about different antenna and their pros and cons

Assume L to be the symbol for λ

YAGI UDA ANTENNA:

This is a type of travelling wave antenna. This has a feed element which is usually a dipole and has a reflector along with other elements which help transmit and reflect the waves in a certain direction. Typically has a gain of about 10dB. For a spacing of 0.3 times the wavelength, a significant drop in gain was noticed. Similar constraints were found on the radii of individual elements. For a separation of $0.2L$, a yagi uda array of 3(reflector, dipole and element) gives a gain of 9, and it increases upon increasing number in the array. However the increase is not linear.

Pros: high gain and has high directivity.

Relatively easy to build. (also available in a PCB form)

Can operate in the Ultra High Frequency range(300 MHz to 3 GHz)

Cons: they experience a degradation of their electrical elements as one moves away from the bandwidth of operating frequency.

DIPOLE ANTENNA:

A dipole antenna is an antenna which is basically two wires running parallel to each other till a point which they turn away and move perpendicular to each other. The wires are connected to the feed element which provides the power to convert into electrical energy by making current flow through the wires.

This is an omni directional antenna with low gain and is usually used in the frequency range of UHF.

The gain of the antenna usually increases from around 1.5 db to 5db for an increase in length up till around $1.5L$

Pros: it is relatively simple to make and use

Omni directional

Can be used in the UHF region

Cons: has a very low gain

Is inefficient

BICONICAL ANTENNA:

This is another omni directional antenna which works in the frequency range of 1GHz to 18GHz(particular model: SAS-547). It has a power consumption of 50 watts and an impedance of 50 ohm. A typical gain is -10 to 4 dBi(gain w.r.t. An isotropic antenna).

Pros: omnidirectional and good gain.

- Relatively small in size

- Has a very wide range of frequencies

Cons: has some characteristics comparable to the dipole antenna in terms of efficiency etc

MICROSTRIP ANTENNA:

This is a directional antenna which is very small and can be printed on a circuit board. It is highly directive and has a good enough gain along with an operating frequency in the microwave region

Pros: highly directive

- Very small in size

- Support multiple bands and polarisation

Cons: very inefficient

- Low gain

- Often radiates from junction points